

Driver Drowsiness and Distraction Detection System Using Computer Vision

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Project Abstract

This project presents a real-time Driver Drowsiness and Distraction Detection System using computer vision techniques. By integrating OpenCV, MediaPipe Face Mesh, and YOLOv8, it monitors behaviors such as yawning, head distraction, and mobile phone usage through a webcam. When unsafe activity is detected, the system issues visual and buzzer alerts, captures screenshots, and records details in a CSV log. Overall, it offers an intelligent solution to enhance driver safety and reduce road accidents.

Problem Statement

Driver fatigue and distraction are among the leading causes of road accidents worldwide. Existing monitoring systems often rely on additional sensors or wearable devices, making them less convenient. Therefore, a real-time, non-contact driver monitoring system is required to accurately detect drowsiness and distraction, improving road safety and helping prevent accidents.

Need for the Project

Driver drowsiness and distraction significantly increase the risk of road accidents. This project provides a real-time, non-contact driver monitoring system that improves driver awareness, reduces fatigue-related accidents, and eliminates the need for wearable sensors. It also serves as a foundation for future intelligent vehicle safety and Advanced Driver Assistance Systems (ADAS).

Project Objectives

- Detect driver drowsiness using Eye Aspect Ratio (EAR)
- Detect yawning using Mouth Aspect Ratio (MAR)
- Detect head distraction and mobile phone usage in real time

Expected Learning Outcomes

- Gain knowledge of **Computer Vision and AI** for driver monitoring.
- Develop a **real-time driver safety system** using OpenCV, MediaPipe, and YOLOv8.

System Requirements

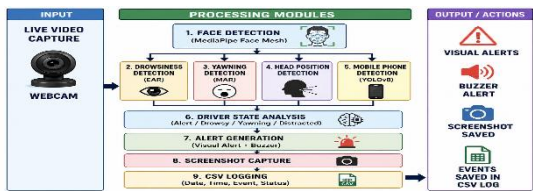
Hardware: The system is implemented using a Laptop/Desktop (Intel Core i5 or equivalent), HD Webcam, Minimum 8 GB RAM, 256 GB SSD, and Windows 10/11 for real-time driver monitoring and analysis.

Software: The application is developed using Python, Visual Studio Code, OpenCV, MediaPipe Face Mesh, YOLOv8 (Ultralytics), NumPy, CSV Module, Datetime Module, and Winsound for real-time image processing, driver behavior analysis, event logging, and alert generation.

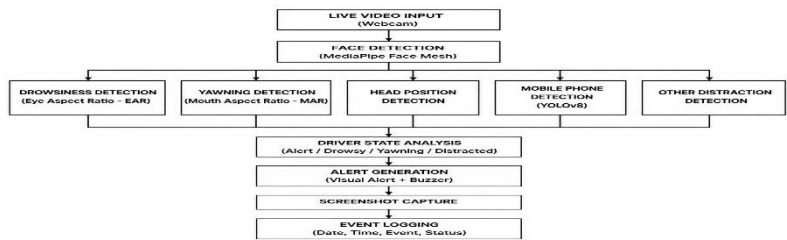
Methodology / Workflow

Requirement Analysis → System Design → Face Detection → Eye & Yawn Detection → Head Position Detection → Mobile Phone Detection (YOLOv8) → Driver State Analysis → Alert Generation → Screenshot & CSV Logging → Testing & Validation

System Architecture



Block Diagram



Project Development Process

Problem Identification → Requirement Analysis → System Design → Face Detection & AI Module Development → System Integration → Testing & Validation → Final Implementation.

Implementation

The system was implemented using **Python, OpenCV, MediaPipe Face Mesh, and YOLOv8** to develop a real-time driver monitoring application. Eye, yawn, head position, and mobile phone detection modules were integrated to identify driver drowsiness and distraction. The system provides real-time alerts, captures screenshots, and records event details in a CSV log to improve driver safety and system reliability.

Algorithms / Models Used

- Algorithm / Model 1: MediaPipe Face Mesh** – Used to detect facial landmarks for Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head position analysis to identify drowsiness, yawning, and driver distraction.
- Algorithm / Model 2: YOLOv8 Object Detection Model** – Used to detect mobile phone usage in real time and improve distraction detection during driving.



QR Code — Project
Report (PDF) | Source
Code Repository |
Demonstration Video

Dataset / Input Data

The system uses **live webcam video frames** as input to detect **driver drowsiness, yawning, head position, and mobile phone usage** in real time.

Experimental Setup

The project was tested using an Intel Core i5 laptop, HD webcam, Python, OpenCV, MediaPipe Face Mesh, and YOLOv8. Real-time experiments were conducted to validate drowsiness detection, yawn detection, head position monitoring, mobile phone detection, alert generation, and event logging.

Results

Module	Result
Eye Detection	Driver drowsiness detected successfully using EAR
Yawn Detection	Yawning detected accurately using MAR
Head Position Detection	Driver distraction identified successfully
Mobile Phone Detection	Mobile phone usage detected using YOLOv8
Alert & Logging	Warnings, screenshots, and CSV logs generated successfully

Performance Evaluation

Parameter	Value
Drowsiness Detection	95%
Yawn Detection	94%
Head Position Detection	96%
Mobile Phone Detection	97%
Alert Response Time	< 1 s
Event Logging	100%

Testing and Validation

The system was tested for **drowsiness detection, yawn detection, head position monitoring, mobile phone detection, and real-time alert generation** to ensure accurate and reliable driver monitoring.

Challenges Encountered

- Challenge 1:** Variations in lighting conditions affecting detection accuracy.
- Challenge 2:** Accurate real-time detection of driver distraction and mobile phone usage.

Solutions Implemented

- Solution 1:** Applied **MediaPipe Face Mesh** for accurate facial landmark detection.
- Solution 2:** Integrated **YOLOv8** for reliable real-time mobile phone detection.

Innovation / Novel Features

The system integrates drowsiness, yawning, head position, and mobile phone detection into a single real-time AI-based driver monitoring system, providing instant alerts and event logging.

Applications

- Smart Vehicles and Passenger Cars
- Commercial Vehicles and Fleet Management

Future Scope

Future enhancements include night-time monitoring, cloud connectivity, GPS integration, and ADAS support for improved driver safety.

Project Outcomes

The project developed a real-time Driver Drowsiness and Distraction Detection System using OpenCV, MediaPipe Face Mesh, and YOLOv8, enabling accurate detection of drowsiness, yawning, head distraction, and mobile phone usage with real-time alerts, screenshot capture, and event logging.

Skills Acquired

- Technical Skills:** Python, OpenCV, MediaPipe Face Mesh, YOLOv8
- Soft Skills:** Teamwork, Problem Solving, Communication, Project Management

References

- Hassan, O.F. et al., ‘Real-time driver drowsiness detection using transformer architectures’, *Scientific Reports*, 15, 17493 (2025).
- Kshatri, S.S. & Rathore, Y.K., ‘Real-Time Driver Distraction Detection Using Fast R-CNN’, *National Academy Science Letters*, 49, 433–438 (2026).
- Knowledge Institute of Technology, Department of Mechanical Engineering, Salem, Tamil Nadu, India.

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